VENKATKUMAR RAJAN

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PROFESSIONAL SUMMARY

Generative AI Engineer with **3.5+ years** of experience designing and deploying production-grade AI systems across multimodal applications. Proven ability to translate cutting-edge research into scalable, real-world solutions that deliver measurable business impact, with a strong focus on secure, responsible AI systems and governance. **Kaggle Master** and author of open-source LLM evaluation library **llmevalkit**.

EXPERIENCE

Generative AI Engineer | *Business Analyst* **Dec 2023 – Present**
EXL Service (India) Pvt. Ltd. *Noida, India (Remote)*

- **AI-Powered Credential Extraction System** *(Unified Text + Multimodal RAG, Azure):* Architected and deployed a production-grade, config-driven per-field RAG pipeline for automated extraction of **50+ structured fields** (single and multi-value) from complex, unstructured medical credentialing documents (PDFs, scanned images, RTF, DOCX, mixed formats). Designed a doctype-agnostic system enabling new document types through prompt and configuration changes without code modifications. Leveraged Azure OpenAI (GPT-5) with dual API support (real-time + batch for cost optimization), Azure Document Intelligence (OCR, layout parsing), Azure AI Search (vector indexing with optional semantic reranking), and embedding pipelines for per-field retrieval. Implemented parallel extraction with ThreadPoolExecutor (configurable workers with retry/backoff), frequency-based aggregation with case-insensitive deduplication, and confidence-based routing (high/low confidence) to improve reliability. Integrated LLM-as-Judge evaluation for field-level scoring, along with provider-level logging, cost tracking, and automated index management. Achieved **80% extraction accuracy** and reduced manual processing effort by **40%+**. Deployed scalable pipelines using Azure DevOps CI/CD, Azure Blob Storage, Key Vault, Azure SQL Server.
- **Unified GenAI Document Intelligence & Automation System** *(Azure):* Built and deployed a real-time, production-grade unified Generative AI document intelligence system for parsing and structuring data across multiple complex document types using Azure OpenAI, NLP, and computer vision. Designed a highly extensible architecture where onboarding new document types requires only a **configuration update and prompt modification**, eliminating redevelopment effort. Achieved **>95% extraction reliability** and reduced manual processing effort by **40–50%**. Deployed scalable pipelines using Azure Blob Storage, Key Vault, Azure DevOps CI/CD, and comprehensive audit logging.
- **AI-Based Document Classification System** *(Unified, Azure):* Built a scalable, **unified** AI document classifier supporting **15+ document categories** with $>$**97% accuracy**. New categories added instantly via a reference document, no engineering or prompt changes required.
- **Custom AI Agent Builder** *(Databricks, Internal):* Designed and built an internal framework enabling users to **create and configure custom AI agents** — connecting multiple Unity Catalog tables, attaching PDF knowledge bases, and defining agent behaviour without writing code. Built specifically to support **denial management analytics**, enabling analysts to perform natural-language queries, root-cause analysis, and insight generation over structured and unstructured data. Deployed as a governed, reusable Databricks App with RAG-powered responses.
- **Internal AI Productivity Tools:** Built a suite of internal tools to accelerate team workflows — **Excel AI assistant** (natural-language queries over spreadsheets), **SQL chatbot** (plain-English to SQL), **Power App** integrations, and **Document Intelligence field extraction** for automated form processing — reducing manual effort across data, operations, and analyst teams.
- Delivered **10+ GenAI PoC presentations** to prospective clients; broad experience across AI system design, enterprise data pipelines, and production LLM deployment.

Machine Learning Engineer **Sept 2022 – Oct 2023**
AugRay *Chennai, Tamil Nadu (Onsite)*

- **Real-Time Ball Fault Detection** *(Edge AI, Computer Vision):* Designed and deployed a lightweight real-time defect detection system for a leading sports goods manufacturer using MobileNetV2, optimised for NVIDIA Jetson AGX. Achieved **90%+ detection accuracy**, **reduced QA error rates by 40%+**, and **increased manufacturing throughput by 3**$\times$ through efficient edge inference and model optimisation.
- **Automated 3D Reconstruction** *(Generative AI + Photogrammetry + NeRF):* Implemented an end-to-end automated 3D reconstruction pipeline combining traditional photogrammetry with Generative AI and Neural Radiance Fields (NeRF). Achieved **95% reconstruction accuracy** for non-reflective objects and $\sim$**50%** for reflective objects, significantly reducing manual processing time and improving 3D asset scalability.
- **Wall & Floor Segmentation with Intelligent Colour Schemes:** Fine-tuned **SegFormer-B0** (segformer-b0-finetuned-ade-512-512) for wall and floor semantic segmentation, achieving **80% accuracy** and enabling a paint manufacturer to automate interior colour preview generation. **Reduced designer effort by 20+ hours/month** and improved design turnaround time and visualisation consistency.
- **Foot Detection & Virtual Shoe Try-On** *(Computer Vision, AR):* Led development of a CV-based virtual shoe try-on system trained on **10,000+ annotated foot images**. Achieved **79% placement accuracy**, enabling realistic footwear visualisation and contributing to a **5% reduction in customer return rates** during pilot trials.
- **Generative AI Proofs of Concept:** Built multiple GenAI PoCs including automated flyer generation, face texture synthesis, and a site-based conversational chatbot, driving innovation in interactive marketing, customer engagement, and AI-powered digital experiences.

SKILLS

- **Programming Languages:** Python, C++
- **Frameworks & Libraries:** TensorFlow, PyTorch, OpenCV, kornia, Pytorch Lighting scikit-learn, PySpark, NetworkX, Mediapipe, librosa, LangChain, LlamaIndex, gym
- **Web & Backend Frameworks:** Django, GraphQL, FastAPI, Flask, REST APIs, Streamlit, Gradio

- **MLOps & LLMOps:** Model Deployment, CI/CD, Docker, Kubernetes, Kubeflow, MLflow, Kafka, Azure DevOps
- **Cloud & Platforms:** Azure, AWS, Databricks
- **Edge AI:** NVIDIA Jetson AGX, Raspberry Pi
- **Database, Automation & Visualization:** MongoDB, MySQL, Power BI, Power Automate, Power Apps, Pytest, Unittest
- **LLM Systems & Evaluation:**LLM Evaluation, AI Agents, Multi-Agent Systems (CrewAI), Retrieval-Augmented Generation (RAG), GraphRAG,Prom Engineering, End-to-End Orchestration
- **Professional Skills:** Leadership, Project Management, Agile Methodologies, Business Analysis, Problem-Solving, Translating Research into Production Systems

PERSONAL PROJECTS & OPEN SOURCE

- **adaptive-intelligence (PyPI v1):** Self-improving RAG orchestration framework using **reinforcement learning** (Thompson Sampling) to dynamically select optimal retrieval strategy per query across 6 routes (vector, keyword, hybrid, graph-augmented, table-first). Features conditional knowledge graph activation (5-signal gate), 6-metric auto-evaluation as RL reward, and adaptive prompt evolution. Achieved **+37% on multi-hop and +25% overall** vs static RAG. Model-agnostic (Ollama, OpenAI, Groq, Grok, HuggingFace), 10+ document formats.
- **llmevalkit (PyPI v6):** LLM evaluation library with **78 metrics across 13 modules** covering quality, hallucination detection, compliance (HIPAA, GDPR, DPDP, EU AI Act), governance (NIST AI RMF, CoSAI, ISO 42001), security, multimodal evaluation, AI content detection, observability, anomaly detection, ground truth testing, conversation evaluation, and red team testing. Supports fully offline and LLM-as-judge modes for LLMs, RAG pipelines, AI agents, and document workflows.
- **Text-to-3D Reconstruction Pipeline** *(Generative AI + NeRF, Research):* Proposed and built a fully automated pipeline that transforms natural language descriptions into detailed 3D models through a multi-stage workflow, beginning with Stable Diffusion–based image generation, followed by RL-agent-driven enhancement, reflection removal using Stable Delight, and subsequent image upscaling and background removal, culminating in volumetric 3D reconstruction. The system addresses challenges in semantic coherence, geometric complexity, and visual fidelity, with applications in AR/VR and digital content creation. Paper
- **PBR Material Classification + Gaussian Splatting** *(Computer Vision, Independent Research):* Independent research on physically-based rendering (PBR) shader principles for interpretable material property extraction from single images — recovering albedo, smoothness, specular, and metallic channels without ground-truth labels. Extended to explore **3D Gaussian Splatting (3DGS)** for novel-view synthesis and single-image multi-view reconstruction. Blog
- **Comprehensive Computer Vision Portfolio** *(Object Detection, Tracking, Video Analysis):* Built a wide range of computer vision solutions spanning object detection, multi-object tracking, and real-time video analysis. Github
- **GenAI RAG/CAG + Knowledge Graph System & Multimodal Chatbot:** Built a Generative AI-powered RAG/CAG and knowledge graph system integrating diverse data sources for efficient information retrieval and context-based generation. Also developed a multimodal chatbot supporting **PDF, CSV, and vision-based drag-and-drop input** with multilingual text + audio responses applicable to real-time domains such as security and drone systems. GitHub
- **Responsible AI Series** (Medium/@VK_Venkatkumar, 6+ parts): Technical articles covering NIST AI RMF, GDPR, CoSAI, HIPAA, LLM Safety Guardrails, Bias Detection, Red Teaming with original architecture diagrams and Python walkthroughs.

RESEARCH & PUBLICATIONS

- A Generative Approach to High Fidelity 3D Reconstruction from Text Data [arXiv] [GitHub]
- Advancing Audio Fingerprinting Accuracy with AI and ML: Addressing Background Noise and Distortion Challenges [IEEE]
- Implementation of PCB Layout using CNC Machine Controlling with Wireless Communication [IRJIET]

CERTIFICATIONS & ACCOMPLISHMENTS

- **Kaggle Master** – Competitions Expert (top ∼0.4% globally), Notebooks Master (top ∼1.3%)
- **Certifications:** TensorFlow Developer (Google), Nvidia Jetson AI Specialist, Microsoft Azure AI, Self-Driving Car Engineer (Udacity Nanodegree), Reinforcement Learning, Computer Vision, Generative AI, AI for Cybersecurity, AI Security

EDUCATION

- **M.Tech. in Artificial Intelligence & Machine Learning** — **Rajasthan, India**
  Birla Institute of Technology and Science (BITS), Pilani — *Mar 2023 – Apr 2025*
- **B.E. in Electronics and Communication Engineering** — **Madurai, Tamil Nadu, India**
  SACS MAVMM Engineering College, Anna University — *Jul 2016 – Oct 2020*